



M-Tech Production & Marketing

Technology | Training | Marketing | Projects

Week 3 - Day 1 Report

Internship Program — Batch 2026

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1. Introduction

This report summarizes the activities completed during the **M-Tech Production & Marketing AI/ML Internship Program** on **12 July 2026**. During today's session, I successfully studied and completed **Experiment 09: Deep Learning – Neural Networks & Backpropagation** from the **Artificial Intelligence & Machine Learning Lab Manual**. The experiment introduced the fundamental concepts of **Deep Learning** and explained how **Artificial Neural Networks (ANNs)** imitate the learning process of the human brain to recognize patterns, process information, and solve complex problems. I also learned about **Backpropagation**, the core training algorithm that enables neural networks to improve their predictions by minimizing errors during the learning process.

2. Objectives

The primary objectives of today's learning session were:

- To understand the fundamental concepts of **Deep Learning**.
 - To study the architecture and working principles of **Artificial Neural Networks (ANNs)**.
 - To learn the roles of **neurons, weights, biases, and hidden layers** in neural networks.
 - To understand the **forward propagation** and **backpropagation** processes.
 - To explore the importance of **activation functions** in solving complex problems.
 - To understand the practical applications of neural networks in modern Artificial Intelligence systems.
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3. Task Description

Today's task involved studying **Experiment 09** from the AI/ML Lab Manual, which focused on the fundamentals of **Deep Learning** and the architecture of **Artificial Neural Networks (ANNs)**.

The experiment explained how input data flows through multiple layers of neurons, how predictions are generated during **forward propagation**, and how prediction errors are calculated by comparing the model's output with the expected results. I also learned how the **Backpropagation** algorithm adjusts the network's weights and biases to minimize prediction errors and continuously improve model accuracy during training.

Additionally, I studied the role of **activation functions** such as **ReLU**, **Sigmoid**, and **Softmax**, which enable neural networks to model complex, non-linear relationships in data.

4. 4. Work Done

During today's internship session, I successfully completed **Experiment 09** from the AI/ML Lab Manual.

I began by studying the fundamentals of **Deep Learning** and understanding how **Artificial Neural Networks** are inspired by the structure and functioning of the human brain. I explored the basic building blocks of neural networks, including **neurons**, **weights**, **biases**, **input layers**, **hidden layers**, and **output layers**, and learned how these components work together to process information.

Next, I studied the concept of **forward propagation**, where input data passes through the network layer by layer until an output prediction is generated. I learned how each neuron performs mathematical computations using weights, biases, and activation functions to produce meaningful outputs.

I then explored the **Backpropagation** algorithm, which calculates prediction errors and propagates them backward through the network to update weights and biases. This iterative optimization process enables the neural network to gradually reduce errors and improve prediction accuracy with each training cycle.

Furthermore, I studied several commonly used **activation functions**, including **ReLU (Rectified Linear Unit)**, **Sigmoid**, and **Softmax**, and understood their individual roles in enabling neural networks to solve complex non-linear problems.

Finally, I explored the practical applications of Deep Learning in various domains, including **computer vision**, **natural language processing (NLP)**, **speech recognition**, **healthcare**, **robotics**, and **autonomous systems**. These examples demonstrated the significant impact of Deep Learning in solving real-world Artificial Intelligence problems.

5. 5. Challenges Faced

The concepts presented in **Experiment 09** were both interesting and informative. However, understanding the mathematical process involved in **Backpropagation**, particularly the calculation of gradients and optimization of weights, required additional attention and practice.

Distinguishing between different activation functions and understanding when each should be used also required careful study. By reviewing theoretical concepts, analyzing practical examples, and studying the

complete learning process of neural networks, I developed a much stronger understanding of these advanced Deep Learning concepts.

6. 6. Key Learnings

During today's learning session, I achieved the following outcomes:

- Developed a strong understanding of the fundamentals of **Deep Learning**.
- Learned the architecture and working principles of **Artificial Neural Networks (ANNs)**.
- Understood the roles of **neurons, weights, biases, hidden layers, and output layers**.
- Gained practical knowledge of **forward propagation** and **Backpropagation**.
- Studied the importance and applications of **ReLU, Sigmoid, and Softmax** activation functions.
- Understood how neural networks continuously improve prediction accuracy through iterative learning.
- Explored the real-world applications of Deep Learning in **computer vision, natural language processing, healthcare, robotics**, and other AI-driven technologies.

7. 7. Conclusion

The successful completion of **Experiment 09** provided valuable theoretical knowledge and practical insight into **Deep Learning** and **Artificial Neural Networks**. I developed a comprehensive understanding of neural network architecture, the learning process through **forward propagation** and **Backpropagation**, and the importance of activation functions in solving complex prediction problems.

The knowledge gained during today's session has significantly strengthened my understanding of modern Artificial Intelligence techniques and provided a solid foundation for studying more advanced Deep Learning models in future experiments. This learning will greatly support my continued growth in AI and Machine Learning throughout the internship program.

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Field	Details
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Supervisor Name Mr. Mohazin Zahid

Date 13 July, 2026

Field	Details
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Domain	AI/ML Intern
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